explain. They might be the first hints of some previously unsuspected quirk in the nonlinear vibrations of the colliding holes' spacetime curvature. Theorists, intrigued by this prospect, will go back to their simulations and search for signs of such doublet wiggles.

- The 0.03-second-long third segment (of which only the beginning is shown in Figure 10.9) consists of oscillations with fixed frequency and gradually dying amplitude. This is precisely the wave form expected when a deformed black hole pulsates to shake off its deformations, that is, as it *rings down* like a struck bell. The pulsations consist of two dumbbell-type protrusions that circulate around and around the hole's equator and gradually die out as ripples of curvature carry away their energy (Figure 10.2 above).

From the details of the waveforms, the computer extracts not only the history of the inspiral, coalescence, and ringdown; it also extracts the masses and spin rates of the initial holes and the final hole. The initial holes each weighed 25 times what the Sun weighs, and were slowly spinning. The final hole weighs 46 times what the Sun weighs and is spinning at 78 percent of the maximum allowed rate. Four solar masses' worth of energy ($2 \times 25 - 46 = 4$) were converted into ripples of curvature and carried away by the waves. The total surface area of the initial holes was 136,000 square kilometers. The total surface area of the final hole is larger, as demanded by the second law of black-hole mechanics (Chapter 12): 144,000 square kilometers. The waveforms also reveal the distance of the hole from Earth: 1 billion light-years, a result accurate to about 20 percent. The waveforms tell us that we on Earth were looking down nearly perpendicularly onto the plane of the orbit, and are now looking down the north pole of the spinning hole; and they show that the holes' orbit had an eccentricity (elongation) of 30 percent.

The computer determines the holes' location on the sky from the wave crests' times of arrival in Japan, Washington, Louisiana, and Italy. Since Japan was hit first, the holes were more or less overhead in Japan, and underfoot in America and Europe. A detailed analysis of the arrival times gives a best-guess location for the source, and an error box around that location of 1 degree in size. Had the holes been smaller, their waveforms would have oscillated more rapidly and the error box would have been